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Towards Reliable Malaria Diagnosis: Hybrid CNN Framework Based on VGG16 with Data Augmentation and Dropout-Enhanced Training Strategy

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***Abstract*- Plasmodium parasites produce malaria, which may be very dangerous if left untreated. While computer-aided procedures allow for quicker and more accurate detection, conventional diagnostic methods are often expensive and subject to human mistake. Deep learning works especially well for classifying malaria because it makes use of a wealth of visual data. This study employs transfer learning with pre-trained convolutional neural networks (CNNs), such as VGG16, ResNet50, and EfficientNetB3, to differentiate between images of cells infected with malaria and those that are not. On a well-known malaria dataset, VGG16 has the best classification accuracy of 98.15% among them, together with outstanding specificity and sensitivity. To maximize model performance, the approach combines data augmentation with a customized CNN architecture built on the VGG16 architecture, which is improved by dropout regularization along with training callbacks. Numerous tests confirm the efficacy of the suggested approach, offering a dependable and automated way to diagnose malaria.**

***Keywords*—Malaria detection; deep learning; convolutional neural networks (CNNs); transfer learning; VGG16; ResNet50; EfficientNetB3; data augmentation; dropout regularization; probabilistic decision fusion.**

I. INTRODUCTION

Malaria, caused by Plasmodium parasites and transmitted by female Anopheles mosquitoes, remains a serious global health issue with 247 million cases and 619,000 deaths in 2021, mainly affecting children under five [1].

Effective malaria care requires prompt, accurate diagnosis. Conventional methods like rapid diagnostic tests (RDTs) and microscopy remain common. However, microscopy demands skilled staff and equipment, limiting low-resource settings, and yields subjective results [2, 3]. RDTs lack parasite load measurement and their sensitivity varies with Plasmodium type [4].

Deep learning and artificial intelligence (AI) offer promising alternatives. CNNs have demonstrated effectiveness in medical diagnostics, improving malaria screening accuracy and efficacy [5, 6, 7, 8, 9].

This paper presents a deep learning-based malaria diagnosis method using peripheral blood smear images. Pre-trained VGG16, ResNet50, and EfficientNetB3 are evaluated. Data augmentation enhances generalization. VGG16 achieved 98.15%, ResNet50 97.82%, and EfficientNetB3 97.61% accuracy on 5,186 images.

Key contributions of the study include:

- The VGG16 model outperformed ResNet50 (97.82%) and EfficientNetB3 (97.61%), achieving the best classification accuracy (98.15%).
- A thorough data augmentation approach was used to enhance generalization and balance the distribution of classes.
- The approach improved diagnostic reliability by continuously achieving high accuracy across all four malaria subtypes.
- The suggested CNN architecture uses callbacks like EarlyStopping and ReduceLROnPlateau, transfer learning, and dropout regularization to incorporate AI-based screening into clinical hematology.

The study is structured as follows: Section 2 examines diagnostic limitations. Section 3 presents the VGG16-based approach with preprocessing and augmentation. Section 4 details training and assessment. Section 5 analyzes model

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performance, highlighting VGG16's accuracy. Section 6 discusses clinical application for resource-constrained settings.

II. Related Works

A paradigm shift from traditional machine learning to advanced deep learning frameworks has occurred in the categorization of malaria cell images, driven by the demand for precise and scalable diagnostic instruments.

Early machine learning approaches required substantial feature engineering. Vijayalakshmi et al. [10] combined SVM and VGG, achieving 93.1% accuracy but poor precision and limited generalization. Pan et al. [11] found LeNet-5's lightweight design limited feature extraction on complex datasets.

Rajaraman et al. [12] evaluated pre-trained CNNs, including VGG16 and ResNet50, showing feature extraction capability but limited robustness due to inconsistent preprocessing. Cinar and Yildirim [13] found GoogleNet achieved 96.6% accuracy but lacked class imbalance handling and advanced augmentation.

Dong et al. [14] investigated transfer learning models exceeding 95% accuracy but lacked details on improvement and augmentation. Reddy et al. [15] achieved 95.91% with ResNet50 but lacked dataset diversity.

In their evaluation of models like MobileNet-V2 (96.53% accuracy with Adam optimizer), Soylu et al. [16] did not take into account state-of-the-art architectures like EfficientNet or significant augmentation to mitigate overfitting.

Traditional techniques [10] are insufficient for diverse medical images. Deep learning models may overfit on sparse data [11, 13, 15], and computational complexity hinders clinical adoption [9, 13]. Many researchers neglect resource limits, robust augmentation, and practical designs [10, 12, 13, 14, 15].

This work tests VGG16, ResNet50, and EfficientNetB3 using a publicly accessible malaria dataset, achieving accuracies of 98.15%, 97.82%, and 97.61%. The VGG16-based model combines transfer learning, dropout regularization, and thick layers, outperforming previous state-of-the-art results. Adam optimizer, EarlyStopping, and ReduceLROnPlateau optimize training. Multi-operation data augmentation ensures generalization and balanced classes. VGG16 surpasses GoogleNet [13], MobileNet-V2 [16], and ResNet50 [16], offering clinical feasibility and scalability in resource-limited settings.

III. Materials and Methods

A. Material &Method

The suggested malaria screening algorithm uses pre-trained CNNs to classify cell images as parasitized or uninfected. Images are resized to 224×224 pixels and augmented (rotation, flipping, shearing, zooming) to increase diversity. Among VGG16, ResNet50, and EfficientNetB3, VGG16 achieves 98.15% accuracy. Binary cross-entropy, Adam optimizer, dropout, and callbacks reduce overfitting. Performance is verified using accuracy, confusion matrices, and ROC analysis.

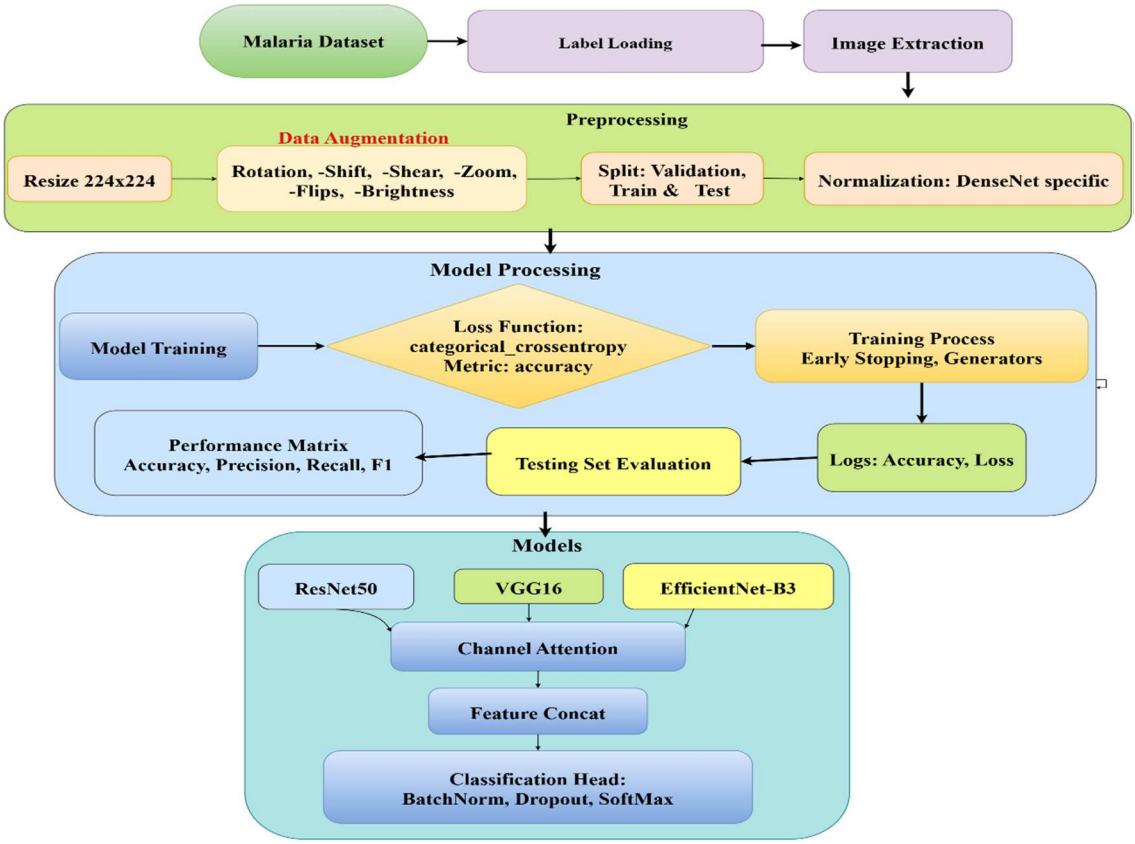


Fig.1: The Proposed Workflow Diagram

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B. Dataset Description

This work employed 27,558 microscopic pictures of red blood cells, evenly split between parasitized and uninfected samples, from the publicly accessible NIH Malaria Dataset [17]. The data was separated into training (70%), validation (15%), and testing (15%) sets using stratified sampling. A 10-fold cross-validation approach was used.

TABLE I. SPLITTING OF THE DATASET

Class	Given Dataset (Train+Test)	Augmented Train	Validation	Test
Parasitized	12,402	11,024	1,378	1,378
Uninfected	12,402	11,024	1,378	1,378

C. Data Preparation and Analysis

The implementation of a common preprocessing pipeline was done to guarantee consistency across all input photos. The following are the crucial steps:

1) Resizing

Malaria photos were downscaled to 224 × 224 pixels using Pillow and ANTIALIAS interpolation to preserve picture integrity, maintain cellular characteristics, and maximize computing efficiency for reliable analysis without sacrificing diagnostic integrity.

2) Normalization

In order to stabilize training and enhance model performance, normalization was used to scale pixel values from [0,255] to [0,1]. This maximizes compatibility with CNN designs by guaranteeing consistency across input data. Images were organized as 4D arrays ((N, Height, Width, Channels)) after normalizing to comply with CNN input specifications.

$$normalized_pixel = \frac{pixel_value}{255.0} \quad (1)$$

Infected cells show malaria parasites as black patches, while parasitized cells appear smooth. The preprocessing pipeline ensures accuracy and data integrity.

3) Data Augmentation

The model's robustness and generalization were enhanced by data augmentation techniques, including rescaling, shear, zoom, vertical and horizontal flips, and Fill Mode to maintain picture integrity during transformations. These additions increase the variety of datasets, reduce overfitting, and enhance convolutional neural networks' (CNNs') ability to detect malaria. The preservation of important parasite traits and the enhancement of variety were balanced by augmentation factors such as zoom and shear ranges (0.2). While larger modifications ran the risk of distorting results, these levels enhanced generality without sacrificing diagnostic precision.

TABLE II. DATA AUGMENTATION STRATEGIES WITH PARAMETER VALUES

Augmentation Strategy	Parameter Value	Description
Rescale	1./255	Normalizes pixel values to the range [0, 1] by dividing by 255.
Shear	0.2	Applies a shear transformation with a maximum shear angle of 20% (0.2 radians).
Zoom	0.2	Randomly zooms in or out by up to 20% of the image size.
Horizontal Flip	True	Randomly flips images horizontally to increase dataset variability.
Vertical Flip	True	Randomly flips images vertically to enhance model robustness.
Fill Mode	'nearest'	Fills newly created pixels after transformations using the nearest pixel value.

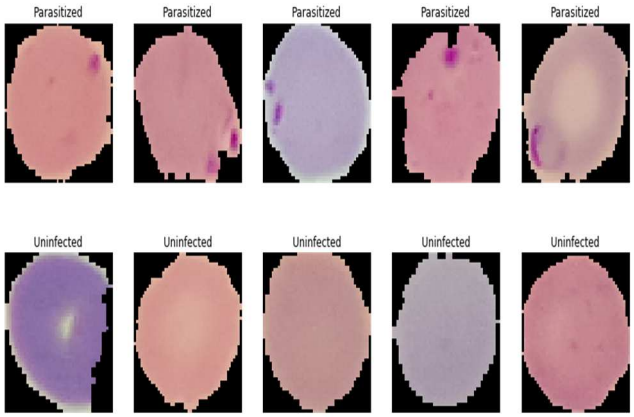


Fig.2. Image After Data Augmentation

D. Hyperparameter Tuning

Using ReduceLROnPlateau (factor=0.2, patience=3, min_lr=1e-6), the model refines the Adam optimizer (learning rate=0.001) to prevent validation loss stagnation. EarlyStopping (patience=5) halts training to avoid overfitting. Dropout layers (0.5, 0.4, 0.3) enhance regularization, improving model generalization, resilience, and malaria detection performance.

TABLE III. HYPERPARAMETER TUNING

Hyperparameter	Value	Description
Optimizer	Adam	Adaptive optimizer with default beta1=0.9, beta2=0.999.
Learning Rate	0.001 (initial)	Initial learning rate, adjusted dynamically.
ReduceLROnPlateau Factor	0.2	Reduces learning rate by 20% if loss plateaus.
ReduceLROnPlateau Patience	3	Waits 3 epochs before reducing learning rate.
ReduceLROnPlateau Min LR	1e-6	Minimum learning rate threshold.
EarlyStopping Patience	5	Stops training after 5 epochs of no improvement.
Epochs	20	Maximum training epochs, may stop early.
Batch Size	32	Standard batch size for ImageDataGenerator.
Dropout Rates	0.5, 0.4, 0.3	Dropout in dense layers (512, 256, 128 units).

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Dense Layer Units	512, 256, 128	Units in custom dense layers with ReLU activation.

E. Fine-Tuning Process

Transfer learning using VGG16, ResNet50, and EfficientNetB3 classifies malaria images, achieving 98.15%, 97.82%, and 97.61% accuracy.

The Adam optimizer maximizes convergence during training by updating model weights according to adaptive learning rates.

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (2)$$

Where,

- θ_t denotes the model parameters at step t
- η is the learning rate (default: 0.001)
- $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected first and second moment estimates
- ϵ is a small constant (typically 10^{-8}) ensuring numerical stability

The following is the definition of the binary cross-entropy loss function, which is employed in the classification task:

Where;

- N is the batch size
- y_i is the true label (1 for Parasitized, 0 for Uninfected)
- $\hat{y}_i$ is the predicted probability for the i-th sample

Class weighting was judged unnecessary due to the dataset's balanced composition (around 13,779 photos per class).

The integration of the following training callbacks improves training convergence and guards against overfitting:

The learning rate is reduced by 0.2 by ReduceLROnPlateau, with a bottom limit of 1×10^{-6} if validation loss remains constant throughout three epochs:

$$\eta_{t+1} = \eta_t \times \text{factor}, \quad \text{subject to } \eta_{t+1} \geq \eta_{\min}$$

EarlyStopping restores the model weights performing the best and stops training if no improvement in validation loss occurs after five consecutive epochs. ImageDataGenerator applies rotation, flipping, shearing, and zooming for real-time data augmentation, enhancing generalization across various blood cell images.

F. Architecture of the VGG16 Transfer Learning Model

This malaria categorization framework is based on the VGG16 model by Simonyan and Zisserman [18]. Pre-trained VGG16 features are used for binary classification—parasitized vs. uninfected cells—via transfer learning. VGG16 was initially trained on ImageNet. We eliminate the top layers (include_top=False) and partly freeze the convolutional base to retain low-level features while fine-tuning higher-level layers.

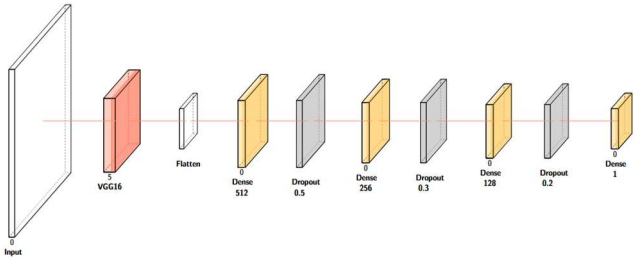


Fig.3: Fine-tuned VGG16 Architecture

Input images are scaled to 224×224×3 and processed through five blocks of convolutional layers with 3x3 filters, stride 1, same padding, max-pooling, and ReLU activation functions.

$$Y_{i,j,k} = ReLU\left(\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{c=0}^{C-1} X_{i+m,j+n,c} \cdot W_{m,n,c,k} + b_k\right) \quad (4)$$

Where,

- $X \in \mathbb{R}^{H\times W\times C}$ is the input tensor,
- $W \in \mathbb{R}^{M\times N\times C\times K}$ is the kernel tensor,
- b_k is the bias term for the k-th filter.

Maximum pooling layers with a 2x2 window and stride 2 reduce spatial dimensions by half to maintain the most dominant activations. After being flattened into a 1D vector of 25,088 features, the last convolutional block produces a (7×7×512) feature map.

The following are included in the custom categorization head:

- A Flatten layer.
- Three fully connected (dense) layers with 512, 256, and 128 units, respectively
- ReLU activation and dropout regularization in each dense layer

The definition of how a dense layer functions is:

$$Z^{(l)} = ReLU\left(W^{(l)} \cdot A^{(l-1)} + b^{(l)}\right) \quad (5)$$

The activation of the preceding layer is denoted by $A^{(l-1)}$, whereas the weight matrix and bias vector are represented by $W^{(l)}$ and $b^{(l)}$. A proportion p of activations are randomly set to zero during training as part of dropout regularization, which is used to reduce overfitting:

$$A_i^{(l)} = \begin{cases} 0 & \text{with probability } p \\ \dfrac{A_i^{(l)}}{1-p} & \text{otherwise} \end{cases} \quad (6)$$

Dropout rates of 0.5, 0.4, and 0.3 are used for the three dense layers respectively.

The last layer generates a probability score for binary classification using a sigmoid activation:

$$\hat{y} = \sigma\left(W^{(L)} \cdot A^{(L-1)} + b^{(L)}\right) = \frac{1}{1 + e^{-\left(W^{(L)} \cdot A^{(L-1)} + b^{(L)}\right)}} \quad (7)$$

A total of 15,141,697 (57.76 MB) parameters 13,406,209 trainable parameters (51.14 MB) Parameters that cannot be trained: 1,735,488 (6.62 MB).

TABLE IV. VGG16 MODEL ARCHITECTURE

Layer (Type)	Output Shape	Parameter Count
VGG16 (Functional)	(None, 512)	14,714,688
Flatten (Flatten)	(None, 512)	0
Dense (Dense)	(None, 512)	262,656

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Dropout (Dropout)	(None, 512)	0
Dense_1 (Dense)	(None, 256)	131,328
Dropout_1 (Dropout)	(None, 256)	0
Dense_2 (Dense)	(None, 128)	32,896
Dropout_2 (Dropout)	(None, 128)	0
Dense_3 (Dense)	(None, 1)	129

IV. Experimental Setup and Performance Metrics

This research compares VGG16, ResNet50, and EfficientNetB3 for classifying malaria images into parasitized and uninfected groups. VGG16 achieves 98.15% accuracy. Augmentation, scaling to 224×224 pixels, and training with Adam optimizer, Dropout, ReduceLROnPlateau, and EarlyStopping improve generalization and prevent overfitting.

A. Performance Evaluation Metrics

The VGG16-based model for malaria classification is evaluated using accuracy, precision, recall, and F1-score to ensure clinical dependability.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (8)$$

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (11)$$

B. Training and Parameter Optimization

In the training process for malaria classification, VGG16, ResNet50, and EfficientNetB3 are used separately; VGG16's accuracy is 98.15%. The dataset of 27,558 images is divided into 70% training and 15% testing. ImageDataGenerator applies augmentation. Adam optimizer improves models. Dropout, EarlyStopping, and ReduceLROnPlateau callbacks ensure convergence.

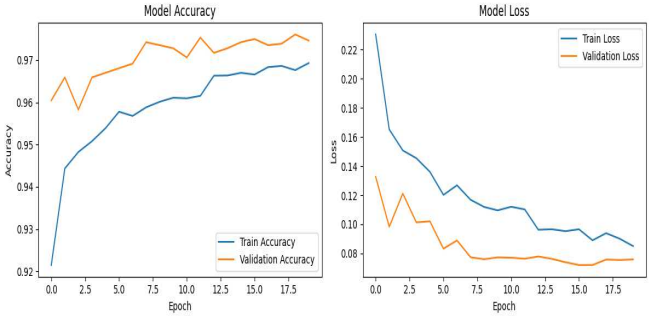


Fig.4: Training and Validation Performance Curves of the VGG16 Framework for Malaria Cell Classification

V. Results Analysis and Discussion

The effectiveness of the VGG16 framework is shown in Table IV, outperforming ResNet50 and EfficientNetB3 with 0.9815 accuracy.This demonstrates how well VGG16 can extract diagnostic characteristics. VGG16 has a 98.15% accuracy rate, however its 15.1 million parameters make it difficult to implement in environments with restricted resources. While VGG16 is appropriate for cloud-based diagnostic systems, lightweight variants such as MobileNet are suitable for portable devices. In addition to data augmentation (rotation, flipping, shearing, zooming) using ImageDataGenerator, dropout (0.5, 0.4, 0.3), EarlyStopping (patience=5), and ReduceLROnPlateau (factor=0.2, patience=3) all help to allay worries about overfitting despite excellent accuracy. Pending further validation, these improvements guarantee versatility across a variety of cell pictures, establishing VGG16 as a solid contender for clinical malaria diagnosis. In borderline situations with concealed parasite traits because of cell overlap or staining problems, misclassifications occurred despite high accuracy, underscoring the need for ensemble approaches or attention mechanisms to improve reliability.

TABLE V. PERFORMANCE COMPARISON OF THE PROPOSED VGG16 ARCHITECTURE

Model	Accuracy	Precision	Recall	F1-Score
VGG16	0.9815	0.9815	0.9815	0.9815
ResNet50	0.9782	0.9782	0.9782	0.9782
EfficientNet-B3	0.9761	0.9761	0.9761	0.9761

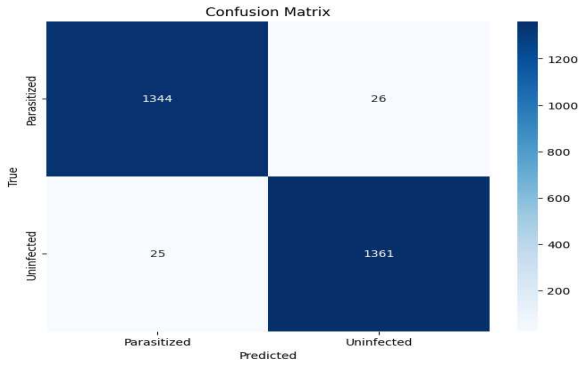


Fig.5.Confusion Matrix

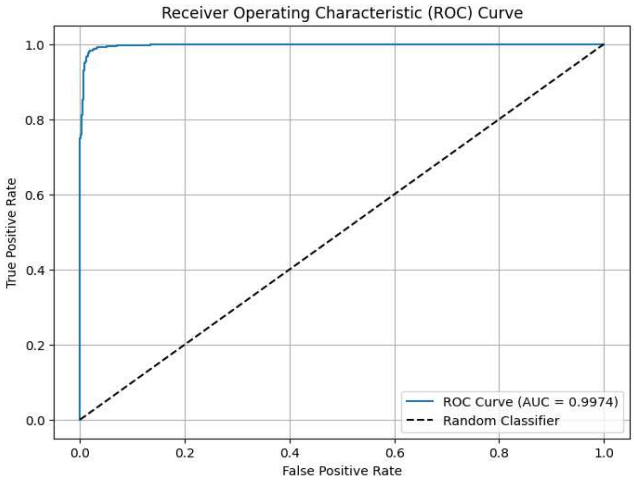


Fig. 6 ROC curve of the proposed VGG16 architecture.

Table VI highlights the shortcomings of current techniques and advantages of the proposed VGG16 framework. Prior works [10], [12], [13], [15], [16] suffer from manual feature engineering, limited augmentation, or imbalance. The proposed

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VGG16 achieves 98.15% accuracy with robust feature extraction and integration.

TABLE VI. COMPARATIVE THE PERFORMANCE OF VGG16 VS EXISTING CLASSIFIERS

Source	Model	Images	Accuracy
Vijayalakshmi et al. [7]	VGG+SVM	2,550	93.10
Rajaraman et al. [9]	ResNet-50	27,558	95.90
Cinar et al. [10]	DenseNet-201	6,730	97.83
Reddy et al. [12]	ResNet50	27,558	95.91
Soylu et al. [13]	MobileNet-v2	27,558	96.53
Proposed model	VGG16	27,558	98.15

V. CONCLUSIONS

The proposed VGG16-based system achieves 98.15% accuracy in malaria recognition, outperforming ResNet50 (97.82%) and EfficientNetB3 (97.61%) in the NIH Malaria Dataset. Transfer learning, data augmentation, and robust regularization ensure high accuracy while overcoming manual feature engineering and dataset imbalance, with computational efficiency suitable for clinical operations. Its promise as a game-changing tool for automated malaria diagnosis will be cemented by further research that confirms its generalizability across other datasets. Through the use of portable devices and telemedicine, an AI model can improve early malaria detection in endemic areas, potentially lowering disease burden and death. Generalizability is constrained, nevertheless, by dependence on the NIH Malaria Dataset. Lightweight, edge-compatible models and validation using a variety of multi-center datasets should be the main topics of future study.

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